**12.** Find integers $x$, $y$, $z$ satisfying

$$\gcd(198, 288, 512) = 198x + 288y + 512z$$

[*Hint:* Put $d = \gcd(198, 288)$. Because $\gcd(198, 288, 512) = \gcd(d, 512)$, first find integers $u$ and $v$ for which $\gcd(d, 512) = du + 512v$.]

## 2.5 THE DIOPHANTINE EQUATION $ax + by = c$

We now change focus somewhat and take up the study of Diophantine equations. The name honors the mathematician Diophantus, who initiated the study of such equations. Practically nothing is known of Diophantus as an individual, save that he lived in Alexandria sometime around 250 A.D. The only positive evidence as to the date of his activity is that the Bishop of Laodicea, who began his episcopate in 270, dedicated a book on Egyptian computation to his friend Diophantus. Although Diophantus' works were written in Greek and he displayed the Greek genius for theoretical abstraction, he was most likely a Hellenized Babylonian. The only personal particulars we have of his career come from the wording of an epigram-problem (apparently dating from the 4th century): His boyhood lasted 1/6 of his life; his beard grew after 1/12 more; after 1/7 more he married, and his son was born 5 years later; the son lived to half his father's age and the father died 4 years after his son. If $x$ was the age at which Diophantus died, these data lead to the equation

$$\frac{1}{6}x + \frac{1}{12}x + \frac{1}{7}x + 5 + \frac{1}{2}x + 4 = x$$

with solution $x = 84$. Thus, he must have reached an age of 84, but in what year or even in what century is not certain.

The great work upon which the reputation of Diophantus rests is his *Arithmetica*, which may be described as the earliest treatise on algebra. Only six Books of the original thirteen have been preserved. It is in the *Arithmetica* that we find the first systematic use of mathematical notation, although the signs employed are of the nature of abbreviations for words rather than algebraic symbols in the sense with which we use them today. Special symbols are introduced to represent frequently occurring concepts, such as the unknown quantity in an equation and the different powers of the unknown up to the sixth power; Diophantus also had a symbol to express subtraction, and another for equality.

It is customary to apply the term *Diophantine equation* to any equation in one or more unknowns that is to be solved in the integers. The simplest type of Diophantine equation that we shall consider is the linear Diophantine equation in two unknowns:

$$ax + by = c$$

where $a, b, c$ are given integers and $a, b$ are not both zero. A solution of this equation is a pair of integers $x_0$, $y_0$ that, when substituted into the equation, satisfy it; that is, we ask that $ax_0 + by_0 = c$. Curiously enough, the linear equation does not appear in the extant works of Diophantus (the theory required for its solution is to be found in Euclid's *Elements*), possibly because he viewed it as trivial; most of his problems deal with finding squares or cubes with certain properties.

A given linear Diophantine equation can have a number of solutions, as is the case with $3x + 6y = 18$, where

$$3 \cdot 4 + 6 \cdot 1 = 18$$
$$3(-6) + 6 \cdot 6 = 18$$
$$3 \cdot 10 + 6(-2) = 18$$

By contrast, there is no solution to the equation $2x + 10y = 17$. Indeed, the left-hand side is an even integer whatever the choice of $x$ and $y$, whereas the right-hand side is not. Faced with this, it is reasonable to enquire about the circumstances under which a solution is possible and, when a solution does exist, whether we can determine all solutions explicitly.

The condition for solvability is easy to state: the linear Diophantine equation $ax + by = c$ admits a solution if and only if $d \mid c$, where $d = \gcd(a, b)$. We know that there are integers $r$ and $s$ for which $a = dr$ and $b = ds$. If a solution of $ax + by = c$ exists, so that $ax_0 + by_0 = c$ for suitable $x_0$ and $y_0$, then

$$c = ax_0 + by_0 = drx_0 + dsy_0 = d(rx_0 + sy_0)$$

which simply says that $d \mid c$. Conversely, assume that $d \mid c$, say $c = dt$. Using Theorem 2.3, integers $x_0$ and $y_0$ can be found satisfying $d = ax_0 + by_0$. When this relation is multiplied by $t$, we get

$$c = dt = (ax_0 + by_0)t = a(tx_0) + b(ty_0)$$

Hence, the Diophantine equation $ax + by = c$ has $x = tx_0$ and $y = ty_0$ as a particular solution. This proves part of our next theorem.

**Theorem 2.9.** The linear Diophantine equation $ax + by = c$ has a solution if and only if $d \mid c$, where $d = \gcd(a, b)$. If $x_0$, $y_0$ is any particular solution of this equation, then all other solutions are given by

$$x = x_0 + \left(\frac{b}{d}\right)t \qquad y = y_0 - \left(\frac{a}{d}\right)t$$

where $t$ is an arbitrary integer.

***Proof.*** To establish the second assertion of the theorem, let us suppose that a solution $x_0$, $y_0$ of the given equation is known. If $x'$, $y'$ is any other solution, then

$$ax_0 + by_0 = c = ax' + by'$$

which is equivalent to

$$a(x' - x_0) = b(y_0 - y')$$

By the corollary to Theorem 2.4, there exist relatively prime integers $r$ and $s$ such that $a = dr, b = ds$. Substituting these values into the last-written equation and canceling the common factor $d$, we find that

$$r(x' - x_0) = s(y_0 - y')$$

The situation is now this: $r \mid s(y_0 - y')$, with $\gcd(r, s) = 1$. Using Euclid's lemma, it must be the case that $r \mid (y_0 - y')$; or, in other words, $y_0 - y' = rt$ for some integer $t$.

Substituting, we obtain

$$x' - x_0 = st$$

This leads us to the formulas

$$x' = x_0 + st = x_0 + \left(\frac{b}{d}\right)t$$

$$y' = y_0 - rt = y_0 - \left(\frac{a}{d}\right)t$$

It is easy to see that these values satisfy the Diophantine equation, regardless of the choice of the integer $t$; for

$$\begin{aligned} ax' + by' &= a\left[x_0 + \left(\frac{b}{d}\right)t\right] + b\left[y_0 - \left(\frac{a}{d}\right)t\right] \\ &= (ax_0 + by_0) + \left(\frac{ab}{d} - \frac{ab}{d}\right)t \\ &= c + 0 \cdot t \\ &= c \end{aligned}$$

Thus, there are an infinite number of solutions of the given equation, one for each value of $t$.

**Example 2.4.** Consider the linear Diophantine equation

$$172x + 20y = 1000$$

Applying the Euclidean's Algorithm to the evaluation of gcd(172 , 20), we find that

$$\begin{aligned} 172 &= 8 \cdot 20 + 12 \\ 20 &= 1 \cdot 12 + 8 \\ 12 &= 1 \cdot 8 + 4 \\ 8 &= 2 \cdot 4 \end{aligned}$$

whence gcd(172 , 20) = 4. Because 4 | 1000, a solution to this equation exists. To obtain the integer 4 as a linear combination of 172 and 20, we work backward through the previous calculations, as follows:

$$\begin{aligned} 4 &= 12 - 8 \\ &= 12 - (20 - 12) \\ &= 2 \cdot 12 - 20 \\ &= 2(172 - 8 \cdot 20) - 20 \\ &= 2 \cdot 172 + (-17)20 \end{aligned}$$

Upon multiplying this relation by 250, we arrive at

$$\begin{aligned} 1000 = 250 \cdot 4 &= 250[2 \cdot 172 + (-17)20] \\ &= 500 \cdot 172 + (-4250)20 \end{aligned}$$

so that $x = 500$ and $y = -4250$ provide one solution to the Diophantine equation in question. All other solutions are expressed by

$$x = 500 + (20/4)t = 500 + 5t$$
$$y = -4250 - (172/4)t = -4250 - 43t$$

for some integer $t$.

A little further effort produces the solutions in the positive integers, if any happen to exist. For this, $t$ must be chosen to satisfy simultaneously the inequalities

$$5t + 500 > 0 \qquad -43t - 4250 > 0$$

or, what amounts to the same thing,

$$-98\frac{36}{43} > t > -100$$

Because $t$ must be an integer, we are forced to conclude that $t = -99$. Thus, our Diophantine equation has a unique positive solution $x = 5$, $y = 7$ corresponding to the value $t = -99$.

It might be helpful to record the form that Theorem 2.9 takes when the coefficients are relatively prime integers.

**Corollary.** If $\gcd(a, b) = 1$ and if $x_0$, $y_0$ is a particular solution of the linear Diophantine equation $ax + by = c$, then all solutions are given by

$$x = x_0 + bt \qquad y = y_0 - at$$

for integral values of $t$.

Here is an example. The equation $5x + 22y = 18$ has $x_0 = 8$, $y_0 = -1$ as one solution; from the corollary, a complete solution is given by $x = 8 + 22t$, $y = -1 - 5t$ for arbitrary $t$.

Diophantine equations frequently arise when solving certain types of traditional word problems, as evidenced by Example 2.5.

**Example 2.5.** A customer bought a dozen pieces of fruit, apples and oranges, for \$1.32. If an apple costs 3 cents more than an orange and more apples than oranges were purchased, how many pieces of each kind were bought?

To set up this problem as a Diophantine equation, let $x$ be the number of apples and $y$ be the number of oranges purchased; in addition, let $z$ represent the cost (in cents) of an orange. Then the conditions of the problem lead to

$$(z + 3)x + zy = 132$$

or equivalently

$$3x + (x + y)z = 132$$

Because $x + y = 12$, the previous equation may be replaced by

$$3x + 12z = 132$$

which, in turn, simplifies to $x + 4z = 44$.

Stripped of inessentials, the object is to find integers $x$ and $z$ satisfying the Diophantine equation

$$x + 4z = 44 \tag{1}$$

Inasmuch as $\gcd(1, 4) = 1$ is a divisor of 44, there is a solution to this equation. Upon multiplying the relation $1 = 1(-3) + 4 \cdot 1$ by 44 to get

$$44 = 1(-132) + 4 \cdot 44$$

it follows that $x_0 = -132$, $z_0 = 44$ serves as one solution. All other solutions of Eq. (1) are of the form

$$x = -132 + 4t \qquad z = 44 - t$$

where $t$ is an integer.

Not all of the choices for $t$ furnish solutions to the original problem. Only values of $t$ that ensure $12 \geq x > 6$ should be considered. This requires obtaining those values of $t$ such that

$$12 \geq -132 + 4t > 6$$

Now, $12 \geq -132 + 4t$ implies that $t \leq 36$, whereas $-132 + 4t > 6$ gives $t > 34\frac{1}{2}$. The only integral values of $t$ to satisfy both inequalities are $t = 35$ and $t = 36$. Thus, there are two possible purchases: a dozen apples costing 11 cents apiece (the case where $t = 36$), or 8 apples at 12 cents each and 4 oranges at 9 cents each (the case where $t = 35$).

Linear indeterminate problems such as these have a long history, occurring as early as the 1st century in the Chinese mathematical literature. Owing to a lack of algebraic symbolism, they often appeared in the guise of rhetorical puzzles or riddles. The contents of the *Mathematical Classic* of Chang Ch′ iu-chien (6th century) attest to the algebraic abilities of the Chinese scholars. This elaborate treatise contains one of the most famous problems in indeterminate equations, in the sense of transmission to other societies—the problem of the "hundred fowls." The problem states:

> If a cock is worth 5 coins, a hen 3 coins, and three chicks together 1 coin, how many cocks, hens, and chicks, totaling 100, can be bought for 100 coins?

In terms of equations, the problem would be written (if $x$ equals the number of cocks, $y$ the number of hens, $z$ the number of chicks):

$$5x + 3y + \frac{1}{3}z = 100 \qquad x + y + z = 100$$

Eliminating one of the unknowns, we are left with a linear Diophantine equation in the two other unknowns. Specifically, because the quantity $z = 100 - x - y$, we have $5x + 3y + \frac{1}{3}(100 - x - y) = 100$, or

$$7x + 4y = 100$$

This equation has the general solution $x = 4t$, $y = 25 - 7t$, so that $z = 75 + 3t$, where $t$ is an arbitrary integer. Chang himself gave several answers:

$$\begin{array}{lll} x = 4 & y = 18 & z = 78 \\ x = 8 & y = 11 & z = 81 \\ x = 12 & y = 4 & z = 84 \end{array}$$

A little further effort produces all solutions in the positive integers. For this, $t$ must be chosen to satisfy simultaneously the inequalities

$$4t > 0 \qquad 25 - 7t > 0 \qquad 75 + 3t > 0$$

The last two of these are equivalent to the requirement $-25 < t < 3\frac{4}{7}$. Because $t$ must have a positive value, we conclude that $t = 1, 2, 3$, leading to precisely the values Chang obtained.

## PROBLEMS 2.5

**1.** Which of the following Diophantine equations cannot be solved?
   (a) $6x + 51y = 22$.
   (b) $33x + 14y = 115$.
   (c) $14x + 35y = 93$.

**2.** Determine all solutions in the integers of the following Diophantine equations:
   (a) $56x + 72y = 40$.
   (b) $24x + 138y = 18$.
   (c) $221x + 35y = 11$.

**3.** Determine all solutions in the positive integers of the following Diophantine equations:
   (a) $18x + 5y = 48$.
   (b) $54x + 21y = 906$.
   (c) $123x + 360y = 99$.
   (d) $158x - 57y = 7$.

**4.** If $a$ and $b$ are relatively prime positive integers, prove that the Diophantine equation $ax - by = c$ has infinitely many solutions in the positive integers.
   [*Hint:* There exist integers $x_0$ and $y_0$ such that $ax_0 + by_0 = c$. For any integer $t$, which is larger than both $|x_0|/b$ and $|y_0|/a$, a positive solution of the given equation is $x = x_0 + bt$, $y = -(y_0 - at)$.]

**5.** (a) A man has \$4.55 in change composed entirely of dimes and quarters. What are the maximum and minimum number of coins that he can have? Is it possible for the number of dimes to equal the number of quarters?
   (b) The neighborhood theater charges \$1.80 for adult admissions and \$.75 for children. On a particular evening the total receipts were \$90. Assuming that more adults than children were present, how many people attended?
   (c) A certain number of sixes and nines is added to give a sum of 126; if the number of sixes and nines is interchanged, the new sum is 114. How many of each were there originally?

**6.** A farmer purchased 100 head of livestock for a total cost of \$4000. Prices were as follow: calves, \$120 each; lambs, \$50 each; piglets, \$25 each. If the farmer obtained at least one animal of each type, how many of each did he buy?

**7.** When Mr. Smith cashed a check at his bank, the teller mistook the number of cents for the number of dollars and vice versa. Unaware of this, Mr. Smith spent 68 cents and then

noticed to his surprise that he had twice the amount of the original check. Determine the smallest value for which the check could have been written.
[*Hint:* If $x$ denotes the number of dollars and $y$ the number of cents in the check, then $100y + x - 68 = 2(100x + y)$.]

**8.** Solve each of the puzzle-problems below:

(a) Alcuin of York, 775. One hundred bushels of grain are distributed among 100 persons in such a way that each man receives 3 bushels, each woman 2 bushels, and each child $\frac{1}{2}$ bushel. How many men, women, and children are there?

(b) Mahaviracarya, 850. There were 63 equal piles of plantain fruit put together and 7 single fruits. They were divided evenly among 23 travelers. What is the number of fruits in each pile?
[*Hint:* Consider the Diophantine equation $63x + 7 = 23y$.]

(c) Yen Kung, 1372. We have an unknown number of coins. If you make 77 strings of them, you are 50 coins short; but if you make 78 strings, it is exact. How many coins are there?
[*Hint:* If $N$ is the number of coins, then $N = 77x + 27 = 78y$ for integers $x$ and $y$.]

(d) Christoff Rudolff, 1526. Find the number of men, women, and children in a company of 20 persons if together they pay 20 coins, each man paying 3, each woman 2, and each child $\frac{1}{2}$.

(e) Euler, 1770. Divide 100 into two summands such that one is divisible by 7 and the other by 11.

# CHAPTER 3

# PRIMES AND THEIR DISTRIBUTION

*Mighty are numbers, joined with art resistless.*
EURIPIDES

## 3.1 THE FUNDAMENTAL THEOREM OF ARITHMETIC

Essential to everything discussed herein—in fact, essential to every aspect of number theory—is the notion of a prime number. We have previously observed that any integer $a > 1$ is divisible by $\pm 1$ and $\pm a$; if these exhaust the divisors of $a$, then it is said to be a prime number. In Definition 3.1 we state this somewhat differently.

> **Definition 3.1.** An integer $p > 1$ is called a *prime number*, or simply a *prime*, if its only positive divisors are 1 and $p$. An integer greater than 1 that is not a prime is termed *composite*.

Among the first ten positive integers, 2, 3, 5, 7 are primes and 4, 6, 8, 9, 10 are composite numbers. Note that the integer 2 is the only even prime, and according to our definition the integer 1 plays a special role, being neither prime nor composite.

In the rest of this book, the letters $p$ and $q$ will be reserved, so far as is possible, for primes.

Proposition 14 of Book IX of Euclid's *Elements* embodies the result that later became known as the Fundamental Theorem of Arithmetic, namely, that every integer greater than 1 can, except for the order of the factors, be represented as a product of primes in one and only one way. To quote the proposition itself: "If a number be the least that is measured by prime numbers, it will not be measured by any other

prime except those originally measuring it." Because every number $a > 1$ is either a prime or, by the Fundamental Theorem, can be broken down into unique prime factors and no further, the primes serve as the building blocks from which all other integers can be made. Accordingly, the prime numbers have intrigued mathematicians through the ages, and although a number of remarkable theorems relating to their distribution in the sequence of positive integers have been proved, even more remarkable is what remains unproved. The open questions can be counted among the outstanding unsolved problems in all of mathematics.

To begin on a simpler note, we observe that the prime 3 divides the integer 36, where 36 may be written as any one of the products

$$6 \cdot 6 = 9 \cdot 4 = 12 \cdot 3 = 18 \cdot 2$$

In each instance, 3 divides at least one of the factors involved in the product. This is typical of the general situation, the precise result being Theorem 3.1.

**Theorem 3.1.** If $p$ is a prime and $p \mid ab$, then $p \mid a$ or $p \mid b$.

***Proof.*** If $p \mid a$, then we need go no further, so let us assume that $p \nmid a$. Because the only positive divisors of $p$ are 1 and $p$ itself, this implies that $\gcd(p, a) = 1$. (In general, $\gcd(p, a) = p$ or $\gcd(p, a) = 1$ according as $p \mid a$ or $p \nmid a$.) Hence, citing Euclid's lemma, we get $p \mid b$.

This theorem easily extends to products of more than two terms.

**Corollary 1.** If $p$ is a prime and $p \mid a_1 a_2 \cdots a_n$, then $p \mid a_k$ for some $k$, where $1 \leq k \leq n$.

***Proof.*** We proceed by induction on $n$, the number of factors. When $n = 1$, the stated conclusion obviously holds; whereas when $n = 2$, the result is the content of Theorem 3.1. Suppose, as the induction hypothesis, that $n > 2$ and that whenever $p$ divides a product of less than $n$ factors, it divides at least one of the factors. Now let $p \mid a_1 a_2 \cdots a_n$. From Theorem 3.1, either $p \mid a_n$ or $p \mid a_1 a_2 \cdots a_{n-1}$. If $p \mid a_n$, then we are through. As regards the case where $p \mid a_1 a_2 \cdots a_{n-1}$, the induction hypothesis ensures that $p \mid a_k$ for some choice of $k$, with $1 \leq k \leq n - 1$. In any event, $p$ divides one of the integers $a_1, a_2, \ldots, a_n$.

**Corollary 2.** If $p, q_1, q_2, \ldots, q_n$ are all primes and $p \mid q_1 q_2 \cdots q_n$, then $p = q_k$ for some $k$, where $1 \leq k \leq n$.

***Proof.*** By virtue of Corollary 1, we know that $p \mid q_k$ for some $k$, with $1 \leq k \leq n$. Being a prime, $q_k$ is not divisible by any positive integer other than 1 or $q_k$ itself. Because $p > 1$, we are forced to conclude that $p = q_k$.

With this preparation out of the way, we arrive at one of the cornerstones of our development, the Fundamental Theorem of Arithmetic. As indicated earlier, this theorem asserts that every integer greater than 1 can be factored into primes in essentially one way; the linguistic ambiguity *essentially* means that $2 \cdot 3 \cdot 2$ is not considered as being a different factorization of 12 from $2 \cdot 2 \cdot 3$. We state this precisely in Theorem 3.2.

**Theorem 3.2 Fundamental Theorem of Arithmetic.** Every positive integer $n > 1$ can be expressed as a product of primes; this representation is unique, apart from the order in which the factors occur.

***Proof.*** Either $n$ is a prime or it is composite; in the former case, there is nothing more to prove. If $n$ is composite, then there exists an integer $d$ satisfying $d \mid n$ and $1 < d < n$. Among all such integers $d$, choose $p_1$ to be the smallest (this is possible by the Well-Ordering Principle). Then $p_1$ must be a prime number. Otherwise it too would have a divisor $q$ with $1 < q < p_1$; but then $q \mid p_1$ and $p_1 \mid n$ imply that $q \mid n$, which contradicts the choice of $p_1$ as the smallest positive divisor, not equal to 1, of $n$.

We therefore may write $n = p_1 n_1$, where $p_1$ is prime and $1 < n_1 < n$. If $n_1$ happens to be a prime, then we have our representation. In the contrary case, the argument is repeated to produce a second prime number $p_2$ such that $n_1 = p_2 n_2$; that is,

$$n = p_1 p_2 n_2 \qquad 1 < n_2 < n_1$$

If $n_2$ is a prime, then it is not necessary to go further. Otherwise, write $n_2 = p_3 n_3$, with $p_3$ a prime:

$$n = p_1 p_2 p_3 n_3 \qquad 1 < n_3 < n_2$$

The decreasing sequence

$$n > n_1 > n_2 > \cdots > 1$$

cannot continue indefinitely, so that after a finite number of steps $n_{k-1}$ is a prime, call it, $p_k$. This leads to the prime factorization

$$n = p_1 p_2 \cdots p_k$$

To establish the second part of the proof—the uniqueness of the prime factorization—let us suppose that the integer $n$ can be represented as a product of primes in two ways; say,

$$n = p_1 p_2 \cdots p_r = q_1 q_2 \cdots q_s \qquad r \leq s$$

where the $p_i$ and $q_j$ are all primes, written in increasing magnitude so that

$$p_1 \leq p_2 \leq \cdots \leq p_r \qquad q_1 \leq q_2 \leq \cdots \leq q_s$$

Because $p_1 \mid q_1 q_2 \cdots q_s$, Corollary 2 of Theorem 3.1 tells us that $p_1 = q_k$ for some $k$; but then $p_1 \geq q_1$. Similar reasoning gives $q_1 \geq p_1$, whence $p_1 = q_1$. We may cancel this common factor and obtain

$$p_2 p_3 \cdots p_r = q_2 q_3 \cdots q_s$$

Now repeat the process to get $p_2 = q_2$ and, in turn,

$$p_3 p_4 \cdots p_r = q_3 q_4 \cdots q_s$$

Continue in this fashion. If the inequality $r < s$ were to hold, we would eventually arrive at

$$1 = q_{r+1} q_{r+2} \cdots q_s$$

which is absurd, because each $q_j > 1$. Hence, $r = s$ and

$$p_1 = q_1 \qquad p_2 = q_2, \ldots, p_r = q_r$$

making the two factorizations of $n$ identical. The proof is now complete.

Of course, several of the primes that appear in the factorization of a given positive integer may be repeated, as is the case with $360 = 2 \cdot 2 \cdot 2 \cdot 3 \cdot 3 \cdot 5$. By collecting like primes and replacing them by a single factor, we can rephrase Theorem 3.2 as a corollary.

**Corollary.** Any positive integer $n > 1$ can be written uniquely in a *canonical form*

$$n = p_1^{k_1} p_2^{k_2} \cdots p_r^{k_r}$$

where, for $i = 1, 2, \ldots, r$, each $k_i$ is a positive integer and each $p_i$ is a prime, with $p_1 < p_2 < \cdots < p_r$.

To illustrate, the canonical form of the integer 360 is $360 = 2^3 \cdot 3^2 \cdot 5$. As further examples we cite

$$4725 = 3^3 \cdot 5^2 \cdot 7 \qquad \text{and} \qquad 17460 = 2^3 \cdot 3^2 \cdot 5 \cdot 7^2$$

Theorem 3.2 should not be taken lightly because number systems do exist in which the factorization into "primes" is not unique. Perhaps the most elemental example is the set $E$ of all positive even integers. Let us agree to call an even integer an $e$-prime if it is not the product of two other even integers. Thus, 2, 6, 10, 14, ... all are $e$-primes, whereas 4, 8, 12, 16, ... are not. It is not difficult to see that the integer 60 can be factored into $e$-primes in two distinct ways; namely,

$$60 = 2 \cdot 30 = 6 \cdot 10$$

Part of the difficulty arises from the fact that Theorem 3.1 is lacking in the set $E$; that is, $6 \mid 2 \cdot 30$, but $6 \nmid 2$ and $6 \nmid 30$.

This is an opportune moment to insert a famous result of Pythagoras. Mathematics as a science began with Pythagoras (569–500 B.C.), and much of the content of Euclid's *Elements* is due to Pythagoras and his School. The Pythagoreans deserve the credit for being the first to classify numbers into odd and even, prime and composite.

**Theorem 3.3 Pythagoras.** The number $\sqrt{2}$ is irrational.

***Proof.*** Suppose, to the contrary, that $\sqrt{2}$ is a rational number, say, $\sqrt{2} = a/b$, where $a$ and $b$ are both integers with $\gcd(a, b) = 1$. Squaring, we get $a^2 = 2b^2$, so that $b \mid a^2$. If $b > 1$, then the Fundamental Theorem of Arithmetic guarantees the existence of a prime $p$ such that $p \mid b$. It follows that $p \mid a^2$ and, by Theorem 3.1, that $p \mid a$; hence, $\gcd(a, b) \geq p$. We therefore arrive at a contradiction, unless $b = 1$. But if this happens, then $a^2 = 2$, which is impossible (we assume that the reader is willing to grant that no integer can be multiplied by itself to give 2). Our supposition that $\sqrt{2}$ is a rational number is untenable, and so $\sqrt{2}$ must be irrational.

There is an interesting variation on the proof of Theorem 3.3. If $\sqrt{2} = a/b$ with $\gcd(a, b) = 1$, there must exist integers $r$ and $s$ satisfying $ar + bs = 1$. As a result,

$$\sqrt{2} = \sqrt{2}(ar + bs) = (\sqrt{2}a)r + (\sqrt{2}b)s = 2br + as$$

This representation of $\sqrt{2}$ leads us to conclude that $\sqrt{2}$ is an integer, an obvious impossibility.

## PROBLEMS 3.1

1. It has been conjectured that there are infinitely many primes of the form $n^2 - 2$. Exhibit five such primes.
2. Give an example to show that the following conjecture is not true: Every positive integer can be written in the form $p + a^2$, where $p$ is either a prime or 1, and $a \geq 0$.
3. Prove each of the assertions below:
   (a) Any prime of the form $3n + 1$ is also of the form $6m + 1$.
   (b) Each integer of the form $3n + 2$ has a prime factor of this form.
   (c) The only prime of the form $n^3 - 1$ is 7.
   [*Hint:* Write $n^3 - 1$ as $(n - 1)(n^2 + n + 1)$.]
   (d) The only prime $p$ for which $3p + 1$ is a perfect square is $p = 5$.
   (e) The only prime of the form $n^2 - 4$ is 5.
4. If $p \geq 5$ is a prime number, show that $p^2 + 2$ is composite.
   [*Hint:* $p$ takes one of the forms $6k + 1$ or $6k + 5$.]
5. (a) Given that $p$ is a prime and $p \mid a^n$, prove that $p^n \mid a^n$.
   (b) If $\gcd(a, b) = p$, a prime, what are the possible values of $\gcd(a^2, b^2)$, $\gcd(a^2, b)$ and $\gcd(a^3, b^2)$?
6. Establish each of the following statements:
   (a) Every integer of the form $n^4 + 4$, with $n > 1$, is composite.
   [*Hint:* Write $n^4 + 4$ as a product of two quadratic factors.]
   (b) If $n > 4$ is composite, then $n$ divides $(n - 1)!$.
   (c) Any integer of the form $8^n + 1$, where $n \geq 1$, is composite.
   [*Hint:* $2^n + 1 \mid 2^{3n} + 1$.]
   (d) Each integer $n > 11$ can be written as the sum of two composite numbers.
   [*Hint:* If $n$ is even, say $n = 2k$, then $n - 6 = 2(k - 3)$; for $n$ odd, consider the integer $n - 9$.]
7. Find all prime numbers that divide 50!.
8. If $p \geq q \geq 5$ and $p$ and $q$ are both primes, prove that $24 \mid p^2 - q^2$.
9. (a) An unanswered question is whether there are infinitely many primes that are 1 more than a power of 2, such as $5 = 2^2 + 1$. Find two more of these primes.
   (b) A more general conjecture is that there exist infinitely many primes of the form $n^2 + 1$; for example, $257 = 16^2 + 1$. Exhibit five more primes of this type.
10. If $p \neq 5$ is an odd prime, prove that either $p^2 - 1$ or $p^2 + 1$ is divisible by 10.
11. Another unproven conjecture is that there are an infinitude of primes that are 1 less than a power of 2, such as $3 = 2^2 - 1$.
    (a) Find four more of these primes.
    (b) If $p = 2^k - 1$ is prime, show that $k$ is an odd integer, except when $k = 2$.
    [*Hint:* $3 \mid 4^n - 1$ for all $n \geq 1$.]
12. Find the prime factorization of the integers 1234, 10140, and 36000.
13. If $n > 1$ is an integer not of the form $6k + 3$, prove that $n^2 + 2^n$ is composite.
    [*Hint:* Show that either 2 or 3 divides $n^2 + 2^n$.]
14. It has been conjectured that every even integer can be written as the difference of two consecutive primes in infinitely many ways. For example,

$$6 = 29 - 23 = 137 - 131 = 599 - 593 = 1019 - 1013 = \cdots$$

    Express the integer 10 as the difference of two consecutive primes in 15 ways.
15. Prove that a positive integer $a > 1$ is a square if and only if in the canonical form of $a$ all the exponents of the primes are even integers.

**16.** An integer is said to be *square-free* if it is not divisible by the square of any integer greater than 1. Prove the following:
   (a) An integer $n > 1$ is square-free if and only if $n$ can be factored into a product of distinct primes.
   (b) Every integer $n > 1$ is the product of a square-free integer and a perfect square. [*Hint:* If $n = p_1^{k_1} p_2^{k_2} \cdots p_s^{k_s}$ is the canonical factorization of $n$, then write $k_i = 2q_i + r_i$ where $r_i = 0$ or 1 according as $k_i$ is even or odd.]

**17.** Verify that any integer $n$ can be expressed as $n = 2^k m$, where $k \geq 0$ and $m$ is an odd integer.

**18.** Numerical evidence makes it plausible that there are infinitely many primes $p$ such that $p + 50$ is also prime. List 15 of these primes.

**19.** A positive integer $n$ is called *square-full*, or *powerful*, if $p^2 \mid n$ for every prime factor $p$ of $n$ (there are 992 square-full numbers less than 250,000). If $n$ is square-full, show that it can be written in the form $n = a^2 b^3$, with $a$ and $b$ positive integers.

## 3.2 THE SIEVE OF ERATOSTHENES

Given a particular integer, how can we determine whether it is prime or composite and, in the latter case, how can we actually find a nontrivial divisor? The most obvious approach consists of successively dividing the integer in question by each of the numbers preceding it; if none of them (except 1) serves as a divisor, then the integer must be prime. Although this method is very simple to describe, it cannot be regarded as useful in practice. For even if one is undaunted by large calculations, the amount of time and work involved may be prohibitive.

There is a property of composite numbers that allows us to reduce materially the necessary computations—but still the process remains cumbersome. If an integer $a > 1$ is composite, then it may be written as $a = bc$, where $1 < b < a$ and $1 < c < a$. Assuming that $b \leq c$, we get $b^2 \leq bc = a$, and so $b \leq \sqrt{a}$. Because $b > 1$, Theorem 3.2 ensures that $b$ has at least one prime factor $p$. Then $p \leq b \leq \sqrt{a}$; furthermore, because $p \mid b$ and $b \mid a$, it follows that $p \mid a$. The point is simply this: A composite number $a$ will always possess a prime divisor $p$ satisfying $p \leq \sqrt{a}$.

In testing the primality of a specific integer $a > 1$, it therefore suffices to divide $a$ by those primes not exceeding $\sqrt{a}$ (presuming, of course, the availability of a list of primes up to $\sqrt{a}$). This may be clarified by considering the integer $a = 509$. Inasmuch as $22 < \sqrt{509} < 23$, we need only try out the primes that are not larger than 22 as possible divisors, namely, the primes 2, 3, 5, 7, 11, 13, 17, 19. Dividing 509 by each of these, in turn, we find that none serves as a divisor of 509. The conclusion is that 509 must be a prime number.

**Example 3.1.** The foregoing technique provides a practical means for determining the canonical form of an integer, say $a = 2093$. Because $45 < \sqrt{2093} < 46$, it is enough to examine the primes 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43. By trial, the first of these to divide 2093 is 7, and $2093 = 7 \cdot 299$. As regards the integer 299, the seven primes that are less than 18 (note that $17 < \sqrt{299} < 18$) are 2, 3, 5, 7, 11, 13, 17. The first prime divisor of 299 is 13 and, carrying out the required division, we obtain $299 = 13 \cdot 23$. But 23 is itself a prime, whence 2093 has exactly three prime factors, 7, 13, and 23:

$$2093 = 7 \cdot 13 \cdot 23$$

Another Greek mathematician whose work in number theory remains significant is Eratosthenes of Cyrene (276–194 B.C.). Although posterity remembers him mainly as the director of the world-famous library at Alexandria, Eratosthenes was gifted in all branches of learning, if not of first rank in any; in his own day, he was nicknamed "Beta" because, it was said, he stood at least second in every field. Perhaps the most impressive feat of Eratosthenes was the accurate measurement of the earth's circumference by a simple application of Euclidean geometry.

We have seen that if an integer $a > 1$ is not divisible by any prime $p \leq \sqrt{a}$, then $a$ is of necessity a prime. Eratosthenes used this fact as the basis of a clever technique, called the *Sieve of Eratosthenes*, for finding all primes below a given integer $n$. The scheme calls for writing down the integers from 2 to $n$ in their natural order and then systematically eliminating all the composite numbers by striking out all multiples $2p, 3p, 4p, 5p, \ldots$ of the primes $p \leq \sqrt{n}$. The integers that are left on the list—those that do not fall through the "sieve"—are primes.

To see an example of how this works, suppose that we wish to find all primes not exceeding 100. Consider the sequence of consecutive integers 2, 3, 4, . . . ,100. Recognizing that 2 is a prime, we begin by crossing out all even integers from our listing, except 2 itself. The first of the remaining integers is 3, which must be a prime. We keep 3, but strike out all higher multiples of 3, so that 9, 15, 21, . . . are now removed (the even multiples of 3 having been removed in the previous step). The smallest integer after 3 that has not yet been deleted is 5. It is not divisible by either 2 or 3—otherwise it would have been crossed out—hence, it is also a prime. All proper multiples of 5 being composite numbers, we next remove 10, 15, 20, . . . (some of these are, of course, already missing), while retaining 5 itself. The first surviving integer 7 is a prime, for it is not divisible by 2, 3, or 5, the only primes that precede it. After eliminating the proper multiples of 7, the largest prime less than $\sqrt{100} = 10$, all composite integers in the sequence 2, 3, 4, . . . ,100 have fallen through the sieve. The positive integers that remain, to wit, 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97, are all of the primes less than 100.

The following table represents the result of the completed sieve. The multiples of 2 are crossed out by \; the multiples of 3 are crossed out by /; the multiples of 5 are crossed out by —; the multiples of 7 are crossed out by ~.

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
| 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
| 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 |
| 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 |
| 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 |
| 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 |
| 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 |
| 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 |
| 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 |
| 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |

By this point, an obvious question must have occurred to the reader. Is there a largest prime number, or do the primes go on forever? The answer is to be found in a remarkably simple proof given by Euclid in Book IX of his *Elements*. Euclid's argument is universally regarded as a model of mathematical elegance. Loosely

speaking, it goes like this: Given any finite list of prime numbers, one can always find a prime not on the list; hence, the number of primes is infinite. The actual details appear below.

**Theorem 3.4 Euclid.** There is an infinite number of primes.

***Proof.*** Euclid's proof is by contradiction. Let $p_1 = 2$, $p_2 = 3$, $p_3 = 5$, $p_4 = 7, \ldots$ be the primes in ascending order, and suppose that there is a last prime, called $p_n$. Now consider the positive integer

$$P = p_1 p_2 \cdots p_n + 1$$

Because $P > 1$, we may put Theorem 3.2 to work once again and conclude that $P$ is divisible by some prime $p$. But $p_1, p_2, \ldots, p_n$ are the only prime numbers, so that $p$ must be equal to one of $p_1, p_2, \ldots, p_n$. Combining the divisibility relation $p \mid p_1 p_2 \cdots p_n$ with $p \mid P$, we arrive at $p \mid P - p_1 p_2 \cdots p_n$ or, equivalently, $p \mid 1$. The only positive divisor of the integer 1 is 1 itself and, because $p > 1$, a contradiction arises. Thus, no finite list of primes is complete, whence the number of primes is infinite.

For a prime $p$, define $p^{\#}$ to be the product of all primes that are less than or equal to $p$. Numbers of the form $p^{\#} + 1$ might be termed *Euclidean numbers*, because they appear in Euclid's scheme for proving the infinitude of primes. It is interesting to note that in forming these integers, the first five, namely,

$$\begin{aligned} 2^{\#} + 1 &= 2 + 1 = 3 \\ 3^{\#} + 1 &= 2 \cdot 3 + 1 = 7 \\ 5^{\#} + 1 &= 2 \cdot 3 \cdot 5 + 1 = 31 \\ 7^{\#} + 1 &= 2 \cdot 3 \cdot 5 \cdot 7 + 1 = 211 \\ 11^{\#} + 1 &= 2 \cdot 3 \cdot 5 \cdot 7 \cdot 11 + 1 = 2311 \end{aligned}$$

are all prime numbers. However,

$$\begin{aligned} 13^{\#} + 1 &= 59 \cdot 509 \\ 17^{\#} + 1 &= 19 \cdot 97 \cdot 277 \\ 19^{\#} + 1 &= 347 \cdot 27953 \end{aligned}$$

are not prime. A question whose answer is not known is whether there are infinitely many primes $p$ for which $p^{\#} + 1$ is also prime. For that matter, are there infinitely many composite $p^{\#} + 1$?

At present, 19 primes of the form $p^{\#} + 1$ have been identified. These correspond to the values $p = 2, 3, 5, 7, 11, 31, 379, 1019, 1021, 2657, 3229, 4547, 4787, 11549, 13649, 18523, 23801, 24029$, and 42209; the largest of these, a number consisting of 18241 digits, was discovered in 2000. The integer $p^{\#} + 1$ is composite for all other $p \leq 120000$.

Euclid's theorem is too important for us to be content with a single proof. Here is a variation in the reasoning: Form the infinite sequence of positive integers

$$\begin{aligned} n_1 &= 2 \\ n_2 &= n_1 + 1 \\ n_3 &= n_1 n_2 + 1 \\ n_4 &= n_1 n_2 n_3 + 1 \\ &\vdots \\ n_k &= n_1 n_2 \cdots n_{k-1} + 1 \\ &\vdots \end{aligned}$$

Because each $n_k > 1$, each of these integers is divisible by a prime. But no two $n_k$ can have the same prime divisor. To see this, let $d = \gcd(n_i, n_k)$ and suppose that $i < k$. Then $d$ divides $n_i$ and, hence, must divide $n_1 n_2 \cdots n_{k-1}$. Because $d \mid n_k$, Theorem 2.2 (g) tells us that $d \mid n_k - n_1 n_2 \cdots n_{k-1}$ or $d \mid 1$. The implication is that $d = 1$, and so the integers $n_k (k = 1, 2, \ldots)$ are pairwise relatively prime. The point we wish to make is that there are as many distinct primes as there are integers $n_k$, namely, infinitely many of them.

Let $p_n$ denote the $n$th of the prime numbers in their natural order. Euclid's proof shows that the expression $p_1 p_2 \cdots p_n + 1$ is divisible by at least one prime. If there are several such prime divisors, then $p_{n+1}$ cannot exceed the smallest of these so that $p_{n+1} \leq p_1 p_2 \cdots p_n + 1$ for $n \geq 1$. Another way of saying the same thing is that

$$p_n \leq p_1 p_2 \cdots p_{n-1} + 1 \qquad n \geq 2$$

With a slight modification of Euclid's reasoning, this inequality can be improved to give

$$p_n \leq p_1 p_2 \cdots p_{n-1} - 1 \qquad n \geq 3$$

For instance, when $n = 5$, this tells us that

$$11 = p_5 \leq 2 \cdot 3 \cdot 5 \cdot 7 - 1 = 209$$

We can see that the estimate is rather extravagant. A sharper limitation on the size of $p_n$ is given by *Bonse's inequality*, which states that

$$p_n^2 < p_1 p_2 \cdots p_{n-1} \qquad n \geq 5$$

This inequality yields $p_5^2 < 210$, or $p_5 \leq 14$. A somewhat better size-estimate for $p_5$ comes from the inequality

$$p_{2n} \leq p_2 p_3 \cdots p_n - 2 \qquad n \geq 3$$

Here, we obtain

$$p_5 < p_6 \leq p_2 p_3 - 2 = 3 \cdot 5 - 2 = 13$$

To approximate the size of $p_n$ from these formulas, it is necessary to know the values of $p_1, p_2, \ldots, p_{n-1}$. For a bound in which the preceding primes do not enter the picture, we have the following theorem.

**Theorem 3.5.** If $p_n$ is the $n$th prime number, then $p_n \leq 2^{2^{n-1}}$.

***Proof.*** Let us proceed by induction on $n$, the asserted inequality being clearly true when $n = 1$. As the hypothesis of the induction, we assume that $n > 1$ and that the result holds for all integers up to $n$. Then

$$\begin{aligned} p_{n+1} &\leq p_1 p_2 \cdots p_n + 1 \\ &\leq 2 \cdot 2^2 \cdots 2^{2^{n-1}} + 1 = 2^{1+2+2^2+\cdots+2^{n-1}} + 1 \end{aligned}$$

Recalling the identity $1 + 2 + 2^2 + \cdots + 2^{n-1} = 2^n - 1$, we obtain

$$p_{n+1} \leq 2^{2^n - 1} + 1$$

However, $1 \leq 2^{2^n - 1}$ for all $n$; whence

$$\begin{aligned} p_{n+1} &\leq 2^{2^n - 1} + 2^{2^n - 1} \\ &= 2 \cdot 2^{2^n - 1} = 2^{2^n} \end{aligned}$$

completing the induction step, and the argument.

There is a corollary to Theorem 3.5 that is of interest.

**Corollary.** For $n \geq 1$, there are at least $n + 1$ primes less than $2^{2^n}$.

***Proof.*** From the theorem, we know that $p_1, p_2, \ldots, p_{n+1}$ are all less than $2^{2^n}$.

We can do considerably better than is indicated by Theorem 3.5. In 1845, Joseph Bertrand conjectured that the prime numbers are well-distributed in the sense that between $n \geq 2$ and $2n$ there is at least one prime. He was unable to establish his conjecture, but verified it for all $n \leq 3{,}000{,}000$. (One way of achieving this is to consider a sequence of primes 3, 5, 7, 13, 23, 43, 83, 163, 317, 631, 1259, 2503, 5003, 9973, 19937, 39869, 79699, 159389, ... each of which is less than twice the preceding.) Because it takes some real effort to substantiate this famous conjecture, let us content ourselves with saying that the first proof was carried out by the Russian mathematician P. L. Tchebycheff in 1852. Granting the result, it is not difficult to show that

$$p_n < 2^n \qquad n \geq 2$$

and as a direct consequence, $p_{n+1} < 2p_n$ for $n \geq 2$. In particular,

$$11 = p_5 < 2 \cdot p_4 = 14$$

To see that $p_n < 2^n$, we argue by induction on $n$. Clearly, $p_2 = 3 < 2^2$, so that the inequality is true here. Now assume that the inequality holds for an integer $n$, whence $p_n < 2^n$. Invoking Bertrand's conjecture, there exists a prime number $p$ satisfying $2^n < p < 2^{n+1}$; that is, $p_n < p$. This immediately leads to the conclusion that $p_{n+1} \leq p < 2^{n+1}$, which completes the induction and the proof.

Primes of special form have been of perennial interest. Among these, the repunit primes are outstanding in their simplicity. A *repunit* is an integer written (in decimal notation) as a string of 1's, such as 11, 111, or 1111. Each such integer must have the form $(10^n - 1)/9$. We use the symbol $R_n$ to denote the repunit consisting of $n$ consecutive 1's. A peculiar feature of these numbers is the apparent scarcity of primes among them. So far, only $R_2$, $R_{19}$, $R_{23}$, $R_{317}$, $R_{1031}$, $R_{49081}$, and $R_{86453}$

have been identified as primes (the last one in 2001). It is known that the only possible repunit primes $R_n$ for all $n \leq 45000$ are the seven numbers just indicated. No conjecture has been made as to the existence of any others. For a repunit $R_n$ to be prime, the subscript $n$ must be a prime; that this is not a sufficient condition is shown by

$$R_5 = 11111 = 41 \cdot 271 \qquad R_7 = 1111111 = 239 \cdot 4649$$

## PROBLEMS 3.2

1. Determine whether the integer 701 is prime by testing all primes $p \leq \sqrt{701}$ as possible divisors. Do the same for the integer 1009.
2. Employing the Sieve of Eratosthenes, obtain all the primes between 100 and 200.
3. Given that $p \nmid n$ for all primes $p \leq \sqrt[3]{n}$, show that $n > 1$ is either a prime or the product of two primes.
   [*Hint:* Assume to the contrary that $n$ contains at least three prime factors.]
4. Establish the following facts:
   (a) $\sqrt{p}$ is irrational for any prime $p$.
   (b) If $a > 0$ and $\sqrt[n]{a}$ is rational, then $\sqrt[n]{a}$ must be an integer.
   (c) For $n \geq 2$, $\sqrt[n]{n}$ is irrational.
   [*Hint:* Use the fact that $2^n > n$.]
5. Show that any composite three-digit number must have a prime factor less than or equal to 31.
6. Fill in any missing details in this sketch of a proof of the infinitude of primes: Assume that there are only finitely many primes, say $p_1, p_2, \ldots, p_n$. Let $A$ be the product of any $r$ of these primes and put $B = p_1 p_2 \cdots p_n / A$. Then each $p_k$ divides either $A$ or $B$, but not both. Because $A + B > 1$, $A + B$ has a prime divisor different from any of the $p_k$, which is a contradiction.
7. Modify Euclid's proof that there are infinitely many primes by assuming the existence of a largest prime $p$ and using the integer $N = p! + 1$ to arrive at a contradiction.
8. Give another proof of the infinitude of primes by assuming that there are only finitely many primes, say $p_1, p_2, \ldots, p_n$, and using the following integer to arrive at a contradiction:

$$N = p_2 p_3 \cdots p_n + p_1 p_3 \cdots p_n + \cdots + p_1 p_2 \cdots p_{n-1}$$

9. (a) Prove that if $n > 2$, then there exists a prime $p$ satisfying $n < p < n!$.
   [*Hint:* If $n! - 1$ is not prime, then it has a prime divisor $p$; and $p \leq n$ implies $p \mid n!$, leading to a contradiction.]
   (b) For $n > 1$, show that every prime divisor of $n! + 1$ is an odd integer that is greater than $n$.
10. Let $q_n$ be the smallest prime that is strictly greater than $P_n = p_1 p_2 \cdots p_n + 1$. It has been conjectured that the difference $q_n - (p_1 p_2 \cdots p_n)$ is always a prime. Confirm this for the first five values of $n$.
11. If $p_n$ denotes the $n$th prime number, put $d_n = p_{n+1} - p_n$. An open question is whether the equation $d_n = d_{n+1}$ has infinitely many solutions. Give five solutions.
12. Assuming that $p_n$ is the $n$th prime number, establish each of the following statements:
   (a) $p_n > 2n - 1$ for $n \geq 5$.
   (b) None of the integers $P_n = p_1 p_2 \cdots p_n + 1$ is a perfect square.
   [*Hint:* Each $P_n$ is of the form $4k + 3$ for $n > 1$.]

(c) The sum

$$\frac{1}{p_1} + \frac{1}{p_2} + \cdots + \frac{1}{p_n}$$

is never an integer.

**13.** For the repunits $R_n$, verify the assertions below:

(a) If $n \mid m$, then $R_n \mid R_m$.

[*Hint:* If $m = kn$, consider the identity

$$x^m - 1 = (x^n - 1)(x^{(k-1)n} + x^{(k-2)n} + \cdots + x^n + 1).]$$

(b) If $d \mid R_n$ and $d \mid R_m$, then $d \mid R_{n+m}$.

[*Hint:* Show that $R_{m+n} = R_n 10^m + R_m$.]

(c) If $\gcd(n, m) = 1$, then $\gcd(R_n, R_m) = 1$.

**14.** Use the previous problem to obtain the prime factors of the repunit $R_{10}$.

## 3.3 THE GOLDBACH CONJECTURE

Although there is an infinitude of primes, their distribution within the positive integers is most mystifying. Repeatedly in their distribution we find hints or, as it were, shadows of a pattern; yet an actual pattern amenable to precise description remains elusive. The difference between consecutive primes can be small, as with the pairs 11 and 13, 17 and 19, or for that matter 1000000000061 and 1000000000063. At the same time there exist arbitrarily long intervals in the sequence of integers that are totally devoid of any primes.

It is an unanswered question whether there are infinitely many pairs of *twin primes*; that is, pairs of successive odd integers $p$ and $p + 2$ that are both primes. Numerical evidence leads us to suspect an affirmative conclusion. Electronic computers have discovered 152892 pairs of twin primes less than 30000000 and 20 pairs between $10^{12}$ and $10^{12}+ 10000$, which hints at their growing scarcity as the positive integers increase in magnitude. Many examples of immense twins are known. The largest twins to date, each 51090 digits long,

$$33218925 \cdot 2^{169690} \pm 1$$

were discovered in 2002.

Consecutive primes cannot only be close together, but also can be far apart; that is, arbitrarily large gaps can occur between consecutive primes. Stated precisely: Given any positive integer $n$, there exist $n$ consecutive integers, all of which are composite. To prove this, we simply need to consider the integers

$$(n + 1)! + 2, (n + 1)! + 3, \ldots, (n + 1)! + (n + 1)$$

where $(n + 1)! = (n + 1) \cdot n \cdots 3 \cdot 2 \cdot 1$. Clearly, there are $n$ integers listed and they are consecutive. What is important is that each integer is composite. Indeed, $(n + 1)! + 2$ is divisible by 2, $(n + 1)! + 3$ is divisible by 3, and so on.

For instance, if a sequence of four consecutive composite integers is desired, then the previous argument produces 122, 123, 124, and 125:

$$5! + 2 = 122 = 2 \cdot 61$$
$$5! + 3 = 123 = 3 \cdot 41$$
$$5! + 4 = 124 = 4 \cdot 31$$
$$5! + 5 = 125 = 5 \cdot 25$$